TONY HE

Software Engineering Intern | Backend Development

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EDUCATION

Carnegie Mellon University | Pittsburgh, PA
Undergraduate Student | Major: Statistics and Machine Learning
Expected Graduation: May 2028

Relevant Coursework: Foundations of Software Engineering, Principles of Imperative Computation, Introduction to Machine Learning, Statistical Computing, Probability and Statistical Inference

TECHNICAL SKILLS

Languages: Python, JavaScript, C, R

Backend & Databases: Node.js, MongoDB, Redis

Developer Tools: Git, GitHub, Docker, Codex, GitHub Actions, Jest, Pytest, Semgrep

Concepts: REST APIs, CI/CD, Automated Testing, Agile Development, LLM Integration, Computer Vision

PROJECT EXPERIENCE

GewuZhizao — AI-Assisted Interior Design Platform

Software Developer | Collaborative Project
September 2026 – Present
GitHub Repository

- Collaborating on an AI-assisted interior design platform that transforms 2D architectural floor plans into interactive, editable 3D interior models.
- Working on computer vision and image processing techniques for automatic wall recognition, structural extraction, and room segmentation from floor plan images.
- Contributing to a design workflow supporting furniture placement, geometric collision detection, material customization, and real-time 3D visualization.
- Supporting iterative development and feature integration using GitHub and Codex, with automated testing and validation of floor plan processing functionality.

NodeBB Course Q&A Enhancement

Backend Developer | CMU Software Engineering Team Project
January 2026 – April 2026

- Collaborated in a four-member development team to enhance the open-source NodeBB discussion platform with course-oriented question management and AI-assisted features.
- Implemented backend API endpoints for marking discussion topics as resolved or unresolved, supporting status-based filtering and automated tests.
- Contributed to an LLM-powered post translation feature using a locally hosted open-source language model integrated with the existing NodeBB architecture.
- Integrated Semgrep static analysis into GitHub Actions and contributed to Docker-based deployment, CI workflows, and backend testing.

Pipedream — Fitbit MCP Integration

Open-Source Contributor | PipedreamHQ/pipedream
April 2026
Merged Pull Request #20666

- Contributed to the open-source Pipedream platform by implementing a Fitbit activity-summary action as part of its MCP tool integration.
- Developed reusable Fitbit API request helpers and structured data retrieval for daily activity metrics, including calories, distance, and active minutes.
- Authored and merged Pull Request #20666, contributing **106 lines of additions across three changed files**.
- Addressed reviewer feedback and performed JavaScript syntax validation to ensure compliance with repository conventions.

ADDITIONAL TECHNICAL EXPERIENCE

Data Structures & Algorithms: Implemented binary search, sorting algorithms, hash dictionaries, stacks, linked lists, and other fundamental data structures using C/C0.

Machine Learning & Statistical Computing: Applied decision trees, classification algorithms, statistical inference, hypothesis testing, and computational data analysis using Python and R.